

OPERATION OMEGA

FINAL SWARM CERTIFICATION AUDIT

ReconPro Enterprise Reconnaissance Platform

Audit Date: August 16, 2026

Version: 0.2.0

Auditor: Independent Swarm (12 Agents)

Protocol: Zero Trust Certification

FINAL VERDICT

NOT READY FOR LAUNCH

CONFIDENTIAL - Evidence-Based Security Assessment

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1. Executive Summary

This report presents the findings of OPERATION OMEGA, a comprehensive independent audit of the ReconPro Enterprise Reconnaissance Platform (v0.2.0). The audit was conducted by 12 specialized engineering agents working independently under a strict Zero Trust protocol. Every finding is backed by direct source code evidence, not marketing claims or prior summaries. The platform suffers from critical security vulnerabilities, broken authentication, complete multi-tenancy data leakage, and a production build that does not compile. The evidence overwhelmingly demonstrates that ReconPro is NOT READY for production launch.

ReconPro is a Next.js 16 application with TypeScript, Prisma ORM, SQLite, and Tailwind CSS. It comprises 56 API routes, 21 Prisma database models, 26 page routes, and approximately 50,000 lines of TypeScript. Despite its ambitions, the platform cannot be deployed to any environment in its current state.

Overall Production Readiness Score

Category	Score	Rating
Architecture	4.2/10	BLOCKED
Security	1.5/10	BLOCKED
Backend APIs	5.0/10	BLOCKED
Frontend/UX	6.0/10	PARTIAL
Database	2.5/10	BLOCKED
Performance	4.0/10	BLOCKED
Deployment	2.0/10	BLOCKED
Scanner Engine	9.5/10	READY
Product Accuracy	5.0/10	PARTIAL

Finding Summary

Severity	Count	Launch Impact
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'CRITICAL' 'frags': [ParaFrag(__tag__='para', bold=1, fontName='DejaVuBd', fontSize=7, greek=0, italic=1, link=[], rise=0, text='CRITICAL', textColor=Color(1,1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '15' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='DejaVu', fontSize=7.5, greek=0, italic=1, link=[], rise=0, text='15', textColor=Color(.898039,.909804,.901961,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Blocks all deployment' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='DejaVu', fontSize=7.5, greek=0, italic=1, link=[], rise=0, text='Blocks all deployment', textColor=Color(.898039,.909804,.901961,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph

Severity	Count	Launch Impact
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'HIGH' 'frags': [P araFrag(__tag__='para' , bold=1, fontName=' DejaVuBd', fontSize=7, greek=0, italic=1, link=[], rise=0, text='HIGH', t extColor=Color(1,1,1, 1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '22' 'frags': [ParaFra g(__tag__='para' , bold=0, fontN ame='DejaVu', fontSize=7.5, greek=0, italic=1, link=[], rise=0, text='22', textC olor=Color(.898 039,.909804,.90 1961,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Must fix before beta' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='DejaVu', fontSize=7.5, greek=0, italic=1, link=[], rise=0, text='Must fix before beta', textColor=Color(.898039,.909804,.901961,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'MEDIUM' 'frags': [ParaFrag(__ta g__='para', bold=1, fo ntName='DejaVuBd', fontSize=7, greek=0, italic=1, link=[], rise=0, text='MEDIUM', textCo lor=Color(1,1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '25' 'frags': [ParaFra g(__tag__='para' , bold=0, fontN ame='DejaVu', fontSize=7.5, greek=0, italic=1, link=[], rise=0, text='25', textC olor=Color(.898 039,.909804,.90 1961,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Should fix pre-launch' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='DejaVu', fontSize=7.5, greek=0, italic=1, link=[], rise=0, text='Should fix pre-launch', textColor=Color(.898039,.909804,.901961,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'LOW' 'frags': [P araFrag(__tag__='para' , bold=1, fontName=' DejaVuBd', fontSize=7, greek=0, italic=1, link=[], rise=0, text='LOW', te xtColor=Color(1,1,1,1) , us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '12' 'frags': [ParaFra g(__tag__='para' , bold=0, fontN ame='DejaVu', fontSize=7.5, greek=0, italic=1, link=[], rise=0, text='12', textC olor=Color(.898 039,.909804,.90 1961,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Technical debt' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='DejaVu', fontSize=7.5, greek=0, italic=1, link=[], rise=0, text='Technical debt', textColor=Color(.898039,.909804,.901961,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph

2. Critical Findings (15 Total)

Each finding is backed by direct source code evidence with file paths and line numbers. These are not theoretical concerns; they are verified vulnerabilities exploitable with trivial effort.

VULN-01: Authentication Bypass - Static Cookie

Location: middleware.ts:20, login/page.tsx:57

Evidence: Cookie "reconpro_auth" value is static "authenticated". Middleware checks only presence. Anyone can bypass all dashboard auth: document.cookie="reconpro_auth=1".

VULN-02: Unsalted SHA-256 Password Hashing

Location: auth/register:82-85, auth/login:58-62

Evidence: Plain SHA-256, no salt, no iterations. GPU cracking at 10B hashes/sec. All identical passwords produce identical hashes.

VULN-03: Unauthenticated Host Reconnaissance

Location: api/system/scan/route.ts:1-73

Evidence: NO auth, NO rate limit. Exposes processes, network interfaces, env var names, SSH config, firewall rules, log contents.

VULN-04: Dev-Mode Auth Bypass

Location: api-protection.ts:198-206

Evidence: DB errors silently bypass auth when NODE_ENV !== production. Staging environments have zero effective auth.

VULN-05: Unauthenticated Scan Proxy

Location: api/scan/stream/route.ts:52-55

Evidence: GET endpoint runs full recon (DNS, SSL, port, HTTP) on any domain without auth. Server abused as anonymized scanning proxy.

VULN-06: Complete Multi-Tenancy Data Leak

Location: scans, dashboard, reports, audit, executive routes

Evidence: Zero routes filter by organizationId. Any user reads all organizations' scans, findings, compliance scores. GDPR violation.

VULN-07: SSRF via HTTP Redirect

Location: safe-fetch.ts:212-214

Evidence: safeFetch validates initial URL only. Redirect to 169.254.169.254 bypasses all SSRF protection when followRedirects=true.

VULN-08: Login Destroys All API Keys

Location: auth/login/route.ts:96-112

Evidence: Every login deactivates ALL active API keys. Breaks all existing integrations and webhooks. No persistent sessions.

BUILD-01: TypeScript Compilation Fails

Location: reports/route.ts:180,186

Evidence: 3 TS errors prevent build. ES2017 target too low for regex dotAll flag and type mismatch in scan route.

BUILD-02: Production Build Fails

Location: next.config.ts + tsconfig.json

Evidence: npm run build fails. No production artifact can be created. ignoreBuildErrors=false prevents broken deployment.

QA-01: Scan Results Never Render

Location: scans/page.tsx:46-65

Evidence: API returns data.scan but client accesses data directly. Every scan shows 0 findings, 0 risk score. Core feature completely broken.

QA-02: Compliance Uses Wrong Org ID

Location: compliance/route.ts:357

Evidence: ScanTarget FK stored as organizationId. Tenant isolation corrupted for all compliance records.

DB-01: Zero Database Indexes

Location: schema.prisma (entire file)

Evidence: Not a single @@index directive. All queries are full table scans. Degradation is linear with data growth.

DB-02: Scan/Finding Lack organizationId

Location: schema.prisma:94-131

Evidence: Cannot filter by tenant without 2-table join through ScanTarget (which itself has nullable organizationId).

PERF-01: Synchronous 20-60s Scan

Location: scan/route.ts:1100-1257

Evidence: Full scan blocks API worker for 20-60+ seconds. No async queue. Causes timeouts and resource exhaustion.

3. High Severity Findings (22 Total)

ID	Finding	Location	Description
H-01	Timing Attack on Password Compare	auth/login:63	Uses !== instead of timingSafeEqual
H-02	API Key in Response Body	auth/login:151	Full key in JSON, logged by proxies
H-03	No Tenant Isolation on Scans	scan/route.ts:1124	No organizationId filter
H-04	Unauthenticated Scans Listing	scans/route.ts:6	All scans returned, no auth
H-05	Dead SSRF Guard Module	ssrf-guard.ts	Canonical guard exists but never imported
H-06	In-Memory Rate Limiting	api-security.ts:183	Per-process Map, bypassable with N instances
H-07	SQLite as Production DB	schema.prisma:7	Single-writer, no concurrent writes
H-08	No Cascade Rules	schema.prisma (all)	Zero onDelete/onUpdate defined
H-09	Stale Auth Headers Hook	use-auth-headers.ts:11	Empty useMemo deps never re-reads localStorage
H-10	Missing useEffect Deps	findings/page.tsx:43	Empty deps cause stale data fetch
H-11	Contact Form Non-Functional	contact-client.tsx:13	Only sets submitted=true
H-12	Forgot Password Not Implemented	forgot-password:65	API succeeds, sends no email
H-13	Compliance Uses All History	compliance:280	No scanId loads ALL findings
H-14	Settings Save Always Fails	settings/page.tsx:62	Missing id in PATCH body
H-15	Settings Loads Wrong Profile	settings/page.tsx:43	Takes members[0] not current user
H-16	Cross-Tenant Audit Leak	integrations/route.ts:103	No org filter on activity log
H-17	No Role-Based Authorization	members/route.ts:155	Viewer can promote self to owner
H-18	No Password Complexity	auth/register:40	Only length >= 8 checked
H-19	DNS Rebinding SSRF	scan/route.ts:1113	IP checked at T1, scan at T2+
H-20	NHI Hardcodes org_default	nhi/route.ts:6	Cross-tenant fallback
H-21	Duplicate ScanTarget IDs	vuln-scan/route.ts:895	Domain-as-ID conflicts with CUID
H-22	No Deployment Mechanism	project root	No Dockerfile, no CI/CD for Next.js

4. API Route Inventory (56 Routes)

Category	Count	%	Details
REAL (Fully Implemented)	28	50%	scan, auth, dashboard, compliance, monitoring, teams, members, integrations, reports, genesis/*, implosion, nhi/*, hall-of-fame, bot-hunter
PARTIAL	11	20%	forgot-password, nhi, nhi/seed, ai-advisor, broadcast/*, sovereign, cni-sentinel, doom-clock, pqc-vault
MOCK/PLACEHOLDER	12	21%	ai-leaderboard, cognitive-dread, exposed-assets, wall-of-shame, fear-index/*, oblivion, sandbox, model-redteam
EMPTY/STUB	1	2%	api/ (root returns "Hello, world")
UNAUTHENTICATED	36	64%	Most GET routes have requireAuth:false
WITH AUTH + ORG FILTER	8	14%	Only monitoring, teams, members, integrations

Only 14% of API routes properly enforce both authentication AND organization-level data isolation. The remaining 86% either lack authentication or fail to filter data by organizationId, enabling cross-tenant data access.

5. Scanner Verification (16/16 Implemented)

All 16 scanner modules were inspected. Every scanner uses real network operations (DNS queries, TLS handshakes, TCP connects, HTTP requests) and produces real findings persisted to the database. This is the strongest aspect of the codebase.

Module	Scanner	Tech
dns-recon.ts	DNS Enumeration	dns/promises, 3 servers, 7 record types
ssl-recon.ts	SSL/TLS Analysis	TLS handshake, cipher/protocol analysis
http-recon.ts	HTTP Header Analysis	Real HTTP, 30+ tech fingerprints
port-check.ts	Port Scanner	TCP connect to 27 ports
whois-recon.ts	WHOIS Intelligence	RDAP HTTPS + TCP WHOIS fallback
subdomain-recon.ts	Subdomain Discovery	DNS perm + crt.sh CT logs
directory-recon.ts	Directory Enumeration	HEAD requests to 80 paths
email-recon.ts	Email Intelligence	MX + SMTP VRFY + SPF/DMARC
cert-recon.ts	Certificate Analysis	CT logs API + TLS cert parsing
geo-recon.ts	Geolocation Mapping	DNS + ip-api.com + reverse DNS
network-recon.ts	Network Interface	os.networkInterfaces() + DNS
process-recon.ts	Process Analyzer	execSync(ps aux) + env scanning
file-recon.ts	File System	fs module, 18 paths, regex secrets
log-recon.ts	Log Analyzer	fs reads /var/log/*, pattern matching
registry-recon.ts	Registry/Config	Reads /etc/* + ufw/iptables
ct-logs.ts	Certificate Transparency	crt.sh API queries

Scanner Engine Score: 9.5/10 - All 16 modules use native Node.js APIs with proper SSRF protection, timeout handling, and result persistence. Production-quality code.

6. Security Certification Matrix (OWASP Top 10)

OWASP Category	Severity	Summary
A01: Broken Access Control	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'CRITICAL' 'frags': [ParaFra g(__tag__='para ' , bold=1, fontN ame='DejaVuBd ' , fontSize=7, greek=0, italic=1, link=[], rise=0, text='CRITICAL', textColor=Color (1,1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	No tenant isolation, unauthenticated data, cookie bypass
A02: Cryptographic Failures	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'CRITICAL' 'frags': [ParaFra g(__tag__='para ' , bold=1, fontN ame='DejaVuBd ' , fontSize=7, greek=0, italic=1, link=[], rise=0, text='CRITICAL', textColor=Color (1,1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Unsalted SHA-256 passwords, API keys without HMAC
A03: Injection	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'LOW' 'frags': [P araFrag(__tag__ ='para', bold=1, fontName='Dej aVuBd', fontSize=7, greek=0, italic=1, link=[], rise=0, text='LOW', tex tColor=Color(1, 1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	No command injection found (native APIs), ReDoS risk

OWASP Category	Severity	Summary
A04: Insecure Design	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'HIGH' 'frags': [P araFrag(__tag__ ='para', bold=1, fontName='Dej aVuBd', fontSize=7, greek=0, italic=1, link=[], rise=0, text='HIGH', tex tColor=Color(1, 1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Session is security theater, no /api/me endpoint
A05: Security Misconfiguration	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'HIGH' 'frags': [P araFrag(__tag__ ='para', bold=1, fontName='Dej aVuBd', fontSize=7, greek=0, italic=1, link=[], rise=0, text='HIGH', tex tColor=Color(1, 1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Dev-mode bypass, deprecated middleware
A06: Vulnerable Components	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'MEDIUM' 'frags': [ParaFra g(__tag__='para ' , bold=1, fontN ame='DejaVuBd ' , fontSize=7, greek=0, italic=1, link=[], rise=0, text='MEDIUM', textColor=Color (1,1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	No known CVEs; prisma in prod deps

OWASP Category	Severity	Summary
A07: Auth Failures	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'CRITICAL' 'frags': [ParaFra g(__tag__='para ' , bold=1, fontN ame='DejaVuBd ' , fontSize=7, greek=0, italic=1, link=[], rise=0, text='CRITICAL', textColor=Color (1,1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Static cookie, timing attack, no brute-force protection
A08: Data Integrity	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'MEDIUM' 'frags': [ParaFra g(__tag__='para ' , bold=1, fontN ame='DejaVuBd ' , fontSize=7, greek=0, italic=1, link=[], rise=0, text='MEDIUM', textColor=Color (1,1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	No audit logging on auth events
A09: Logging Failures	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'MEDIUM' 'frags': [ParaFra g(__tag__='para ' , bold=1, fontN ame='DejaVuBd ' , fontSize=7, greek=0, italic=1, link=[], rise=0, text='MEDIUM', textColor=Color (1,1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Auth events to console.error only

OWASP Category	Severity	Summary
A10: SSRF	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'HIGH' 'frags': [P araFrag(__tag__ ='para', bold=1, fontName='Dej aVuBd', fontSize=7, greek=0, italic=1, link=[], rise=0, text='HIGH', tex tColor=Color(1, 1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Redirect bypass, DNS rebinding, dead ssrf-guard

7. Feature Matrix (Claims vs Reality)

Claimed Feature	Status	Evidence
16 Scanner Modules	Fully Implemented	All 16 use real network operations
PDF Report Generation	MISSING	Only JSON/HTML/Markdown. No PDF code exists
Real-time Monitoring	Partial	CRUD exists, no background scheduler
Team Management	Fully Implemented	Full CRUD at /api/members and /api/teams
Compliance Frameworks	Simulated	Algorithmic estimation, not real audits
CLI Tool	Marketing Only	Terminal demo is hardcoded animation
Integrations (Slack, Jira)	Marketing Only	Zero integration code exists
55 API Endpoints	Approximate	56 routes exist, 66 documented
5 Implemented Scanners	FALSE	Header says 5 but ALL 16 show implemented

8. Performance Review

Issue	Severity	Impact
Zero Database Indexes	Paragraph('caseSensitive': 1, 'encoding': 'utf8', 'text': 'CRITICAL', 'frags': [Paragraph(__tag__='para', bold=1, fontName='DejaVuBd', fontSize=7, greek=0, italic=1, link=[], rise=0, text='CRITICAL', textColor=Color(1,1,1,1), us_lines=[])] 'style': 'bulletText': None, 'debug': 0) #Paragraph	All queries are full table scans
Sequential Finding Inserts	Paragraph('caseSensitive': 1, 'encoding': 'utf8', 'text': 'CRITICAL', 'frags': [Paragraph(__tag__='para', bold=1, fontName='DejaVuBd', fontSize=7, greek=0, italic=1, link=[], rise=0, text='CRITICAL', textColor=Color(1,1,1,1), us_lines=[])] 'style': 'bulletText': None, 'debug': 0) #Paragraph	50-100+ individual INSERTs per scan

Issue	Severity	Impact
Sequential Dashboard Queries	Paragraph('caseSensitive': 1, 'encoding': 'utf8', 'text': 'HIGH', 'frags': [Paragraph(__tag__ = 'para', bold=1, fontName='DejaVuBd', fontSize=7, greek=0, italic=1, link=[], rise=0, text='HIGH', textColor=Color(1,1,1,1), us_lines=[])] 'style': None, 'bulletText': None, 'debug': 0) #Paragraph	10 sequential DB round-trips
Sequential Vuln-Scan Ports	Paragraph('caseSensitive': 1, 'encoding': 'utf8', 'text': 'CRITICAL', 'frags': [Paragraph(__tag__ = 'para', bold=1, fontName='DejaVuBd', fontSize=7, greek=0, italic=1, link=[], rise=0, text='CRITICAL', textColor=Color(1,1,1,1), us_lines=[])] 'style': None, 'bulletText': None, 'debug': 0) #Paragraph	22 ports probed sequentially (132s)

Issue	Severity	Impact
Synchronous 20-60s Scan	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'CRITICAL' 'frags': [Paragraph(__tag__ ='para', bold=1, fontName='DejaVuBd', fontSize=7, greek=0, italic=1, link=[], rise=0, text='CRITICAL', textColor=Color(1,1,1,1) , us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	API route blocked entire duration
No SSR on Landing Page	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'HIGH' 'frags': [Paragraph(__tag__ ='para', bold=1, fontName='DejaVuBd', fontSize=7, greek=0, italic=1, link=[], rise=0, text='HIGH', textColor=Color(1,1,1,1) , us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	All sections ssr:false, blank page

Issue	Severity	Impact
Prisma in Production Deps	<pre>Paragraph(' caseSensitiv e': 1 "encoding': 'utf8' 'text': 'HIGH' 'frags': [Para Frag(__tag__ ='para', bold=1, font Name='Deja VuBd', fontSize=7, greek=0, italic=1, link=[], rise=0, text='HIGH', textColor=C olor(1,1,1,1) , us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>	~10-15MB unnecessary code shipped
N+1 Monitoring Queries	<pre>Paragraph(' caseSensitiv e': 1 "encoding': 'utf8' 'text': 'MEDIUM' 'frags': [Para Frag(__tag__ ='para', bold=1, font Name='Deja VuBd', fontSize=7, greek=0, italic=1, link=[], rise=0, text ='MEDIUM', textColor=C olor(1,1,1,1) , us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>	One DB query per policy

9. Deployment Readiness

Check	Status	Details
TypeScript Compilation	FAIL	3 errors: ES2017 target, type mismatch
ESLint	PASS	Clean pass (30+ rules disabled)
Production Build	FAIL	Fails on TS errors
Prisma Schema	PASS	Validates successfully
Prisma Client	PASS	Generated successfully
Dockerfile	MISSING	No Dockerfile for Next.js app
CI/CD Pipeline	MISSING	No GitHub Actions
Environment Docs	FAIL	No .env.example
Middleware Deprecation	WARNING	Next.js 16 deprecates middleware

10. Launch Checklist

Category	Rating	Explanation
Authentication	BLOCKED	Static cookie bypass, SHA-256 passwords, no sessions
Authorization	BLOCKED	Zero tenant isolation across all data routes
Database	BLOCKED	No indexes, no cascades, SQLite single-writer
API Layer	BLOCKED	64% unauthenticated, scan results broken
Frontend	PARTIAL	Real data fetch but stale auth, broken settings
Scanner Engine	READY	All 16 scanners real and operational
Monitoring	PARTIAL	CRUD works but no background scheduler
Compliance	BLOCKED	Wrong org ID, accumulates history
Reporting	PARTIAL	3 formats work, PDF missing, no auth
Dashboard	PARTIAL	Fetches real data, scans never render
Deployment	BLOCKED	Build fails, no Docker, no CI/CD
Performance	BLOCKED	Zero indexes, sequential ops, 60s scans
Security	BLOCKED	15 critical vulns, 5 OWASP critical
Documentation	PARTIAL	API overview honest, marketing contradicts

11. Top 10 Required Fixes (Priority Order)

ID	Fix	Severity	Effort	Details
FIX-01	Fix tsconfig target ES2018	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'CRITICAL' 'frags': [Paragraph(_tag_='p' 'ara', bold=1, fontName= ='DejaVuBd', fontS ize=7, greek=0, italic=1, link=[], rise=0, text= xt='CRITICAL', text Color=Color(1,1,1,1), us_lines =[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	2m	Change target to ES2018 in tsconfig.json
FIX-02	Fix type mismatch	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'CRITICAL' 'frags': [Paragraph(_tag_='p' 'ara', bold=1, fontName= ='DejaVuBd', fontS ize=7, greek=0, italic=1, link=[], rise=0, text= xt='CRITICAL', text Color=Color(1,1,1,1), us_lines =[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	5m	Change Finding evidence to string null

ID	Fix	Severity	Effort	Details
FIX-03	Real session management	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'CRITICAL' 'frags': [Paragraph(_tag_='p' _tag_='p' _tag_='p' bold=1, fontName='DejaVuBd', fontSize=7, greek=0, italic=1, link=[], rise=0, text='CRITICAL', textColor=Color(1,1,1,1), us_lines=[])]) 'style': 'bulletText': None 'debug': 0) #Paragraph	4h	HttpOnly Secure cookie with random token in DB
FIX-04	Replace SHA-256 with bcrypt	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'CRITICAL' 'frags': [Paragraph(_tag_='p' _tag_='p' _tag_='p' bold=1, fontName='DejaVuBd', fontSize=7, greek=0, italic=1, link=[], rise=0, text='CRITICAL', textColor=Color(1,1,1,1), us_lines=[])]) 'style': 'bulletText': None 'debug': 0) #Paragraph	2h	bcrypt.hash(password, 12) for reg, compare for login

ID	Fix	Severity	Effort	Details
FIX-05	Add orgId to all queries	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'CRITICAL' 'frags': [Paragraph(_tag_='p ara', bold=1, fontName='DejaVu Bd', fontSize=7, greek=0, italic=1, link=[], rise=0, text='CRITICAL', textColor=Color(1,1,1,1), us_lines=[])]) 'style': 'bulletText': None 'debug': 0) #Paragraph	1d	Filter every query by auth.organizationId
FIX-06	Add requireAuth to data routes	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'CRITICAL' 'frags': [Paragraph(_tag_='p ara', bold=1, fontName='DejaVu Bd', fontSize=7, greek=0, italic=1, link=[], rise=0, text='CRITICAL', textColor=Color(1,1,1,1), us_lines=[])]) 'style': 'bulletText': None 'debug': 0) #Paragraph	2h	Set requireAuth:true on 10+ unauthenticated routes

ID	Fix	Severity	Effort	Details
FIX-07	Secure system/scan endpoint	Paragraph('caseSensitive': 1, 'encoding': 'utf8' 'text': 'CRITICAL' 'frags': [Paragraph('_tag_'='p 'ara', bold=1, fontName ='DejaVuBd', fontS ize=7, greek=0, italic=1, link=[], rise=0, text='CRIT ICAL', textColor=Col or(1,1,1,1), us_lines =[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	5m	Add requireAuth + rate limiting
FIX-08	Add auth to scan/stream	Paragraph('caseSensitive': 1, 'encoding': 'utf8' 'text': 'HIGH' 'frags': [Paragraph('_tag_'='p 'ara', bold=1, fontName ='DejaVuBd', fontS ize=7, greek=0, italic=1, link=[], rise=0, text='HIGH' , textColor=Color(1 ,1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	5m	Add requireAuth:true to protection call

ID	Fix	Severity	Effort	Details
FIX-09	Add database indexes	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'HIGH' 'frags': [Paragraph(_tag_='p' _ara', bold=1, fontName= ='DejaVuBd', fontS ize=7, greek=0, italic=1, link=[], rise=0, text='HIGH' , textColor=Color(1 ,1,1,1), underline=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	2h	@@index on all FK and queried columns
FIX-10	Fix scan results rendering	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'HIGH' 'frags': [Paragraph(_tag_='p' _ara', bold=1, fontName= ='DejaVuBd', fontS ize=7, greek=0, italic=1, link=[], rise=0, text='HIGH' , textColor=Color(1 ,1,1,1), underline=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	30m	Destructure data.scan in scans/page.tsx

12-14. Architecture, Database, Contradictions

Architecture Score: 4.2/10 - Clean route groups, proper scanner engine structure, no circular imports. Critical flaws: 1,256-line God route, no service layer, 7 fantasy engine modules (5,226 lines), 41 dead archive components (30K lines), in-memory state breaks horizontal scaling.

Database Score: 2.5/10 - 21 models with appropriate relations. Critical flaws: zero indexes, zero cascade rules, Scan/Finding lack organizationId, nullable ScanTarget.orgId, no migration history, sequential finding inserts. SQLite single-writer prevents concurrent operations.

Contradictions Resolved: The previous summary claimed "only 1 API route exists" - FALSE. 56 route files exist, 28 fully implemented. It claimed "auth sends wrong field names" - FALSE. Fields match. It claimed "5 scanners implemented" - FALSE. All 16 are fully implemented.

15. Recommended Roadmap

Phase 1: Critical Fixes (Week 1)

Fix TypeScript build errors, implement real session management, replace SHA-256 with bcrypt, add organizationId filtering to all routes, add requireAuth to data endpoints, secure/remove system/scan endpoint, fix scan results rendering.

Phase 2: Security Hardening (Weeks 2-3)

Add database indexes, implement cascade rules, fix compliance orgId corruption, add role-based authorization, integrate SSRF guard, add brute-force protection, create Dockerfile and CI/CD pipeline.

Phase 3: Production Readiness (Weeks 4-6)

Migrate SQLite to PostgreSQL, extract scan logic from God routes, implement async scan queue, parallelize DB queries, enable SSR, implement /api/me, remove fantasy endpoints, implement PDF reports, add password reset, batch inserts.

Phase 4: Scale (Weeks 7-10)

Redis rate limiting, TypeScript API client, delete archive components, implement real integrations or remove claims, background monitoring scheduler, migrate middleware to Next.js 16 proxy, WebSocket notifications, integration/e2e tests.

FINAL CERTIFICATION

NOT READY FOR LAUNCH

ReconPro v0.2.0 contains 15 critical vulnerabilities, a broken production build, complete multi-tenancy data leakage, and a fundamentally broken authentication system. Despite having an excellent scanner engine (9.5/10), the platform is NOT suitable for any form of production deployment. Minimum effort to reach closed beta: 4-6 weeks.

Report: 2026-08-16 10:20:19 UTC | OPERATION OMEGA SWARM CERTIFICATION | 12 independent specialists | Zero Trust, Evidence Only